March 18, 2026

Abstract

Continuum robots are a type of soft robot heavily inspired by the motion of elephant trunks. They are able to navigate complex environments through their innate actuation methods and material properties. To achieve fully soft devices, pneumatic actuators are often used, however, the materials used in these actuators are typically delicate and are prone to failure. Recently, an approach to mitigate this is the use of self-healing polymers. This report covers the investigation of a new material based on a polyimide to achieve self-healing capabilities and its use in a soft continuum robot.

Contents

1 Introduction **1**
- 1.1 Importance and Applications of Soft Robotics . . . 1
- 1.2 Types of Self-Healing Polymers . . . 1
- 1.3 Comparison of Different Self-Healing Pathways . . . 4
  - 1.3.1 Using a Diels-Alder Addition as a Self-Healing Pathway . . . 4
  - 1.3.2 Using a Thiol-Michael Addition as a Self-Healing Pathway . . . 5
  - 1.3.3 Using a Pi-Stacking Approach as a Self-Healing Pathway . . . 6
- 1.4 Applications of Self-Healing Polymers in Soft Robotics . . . 6
  - 1.4.1 State of the Art Self-Healing Soft Robots . . . 6
  - 1.4.2 Requirements for Self-Healing Polymers in Soft Robotics . . . 6
- 1.5 Limitations with Current Materials for Soft Robotics . . . 6

2 Project Aims and Objectives **8**

3 Synthesis of a Self-Healing Polymer **9**
- 3.1 Results . . . 9
  - 3.1.1 Hydrolysis of the Polyimide . . . 9
  - 3.1.2 Cross-linking of the Polyimide . . . 9
- 3.2 Hydrolysis of a Polyimide . . . 10
  - 3.2.1 . . . 10
  - 3.2.2 Comparison of Hydrolysed Polyimide versus Purchased Polyamic Acid . . . 10
  - 3.2.3 Spectral Analysis . . . 11
- 3.3 Crosslinking of Polyamic Acid . . . 11
- 3.4 Addition of Diels-Alder Groups . . . 11
- 3.5 Experimental . . . 11
- 3.6 Experimental . . . 11
  - 3.6.1 The Hydrolysis of a Polyimide . . . 11
  - 3.6.2 Crosslinking of Polyamic acid . . . 12
  - 3.6.3 Cross-linking with TEAB and Ethylene carbonate . . . 12
  - 3.6.4 Cross-linking with other salts . . . 12
  - 3.6.5 Spectral Data . . . 12

4 Material Comparison **13**

List of Figures

1.1 A continuum device sustaining damage before self-healing and regaining full functionality using a modified image from[?] 2
1.2 a) The polymer chain with a repeating chemical sub-unit. b) The mechanism of damage and repair for a self-healing polymer 3
1.3 a) A Diels-Alder reaction between two molecules of cylopentadiene. b) A Diels-Alder reaction between a furan and maleimide. c) A Thiol-Michael Addition between a thiol and a conjugated alkene. 5

Chapter 1

Introduction

1.1 Importance and Applications of Soft Robotics

Soft robots are a type of robot known for their use of soft materials, compliant nature and near infinite degrees of freedom . Degrees of freedom (DOF) is a term related to the number of independent movements a robot can make in 3D space.[?] Soft robots are usually nature inspired such as "vine-like" tip growing robots,[?,?] "snake like" continuum robots[?,?] and "hand-like" soft grippers.[?,?] The advantages of using a body that is highly deformable and with near-infinite DOF means these devices can navigate tortuous environments with ease.

This report focuses on continuum robots, often characterised by their long flexible bodies, commonly used for navigation and delivery or retrieval of objects in constrained environments.[?] Continuum robots can be classified into "continuous backbone" and "segmented back-bone" designs.[?] Continuous backbone devices are made from a soft and compliant material, in this report we will call these "soft-bodied", whereas, a segmented backbone refers to those that are made from hard, inflexible segments that are flexibly attached to achieve high curvatures, referred to here as "hard-bodied".

Hard-bodied continuum robots benefit from stiffer materials allowing a greater ease of control[?] which can improve dexterity. However, they cannot deform and are more likely to impart a greater force on the surrounding environment.[?] Hard continuum robots are often tendon driven, whereby the segments are joined by wires that can be pulled to give specific movements. Whereas soft bodied continuum robots can achieve high curvatures and low forces on surroundings,[?] but are often harder to control due to their methods of actuation[?] and can suffer from material failure.[?] Soft continuum robots commonly take the form of an inflated body, these can be controlled in many ways such as varying pressure,[?] strain limiting a specific region[?,?] or magnetic actuation.[?,?] Pressurised systems that rely on fluids need a constant seal to maintain function as opposed to a segmented tendon driven approach that can be partially damaged and still actuatable. This leads the development of resilient materials such as self-healing polymers that can be applied to soft robotics.

1.2 Types of Self-Healing Polymers

Polymers are a type of molecule that consists of several repeating sub-units, these form chains that are combined many times over to form materials. Self-healing (S-H) polymers are a type of polymer that can heal damage.[?] This occurs through reactions and interac-

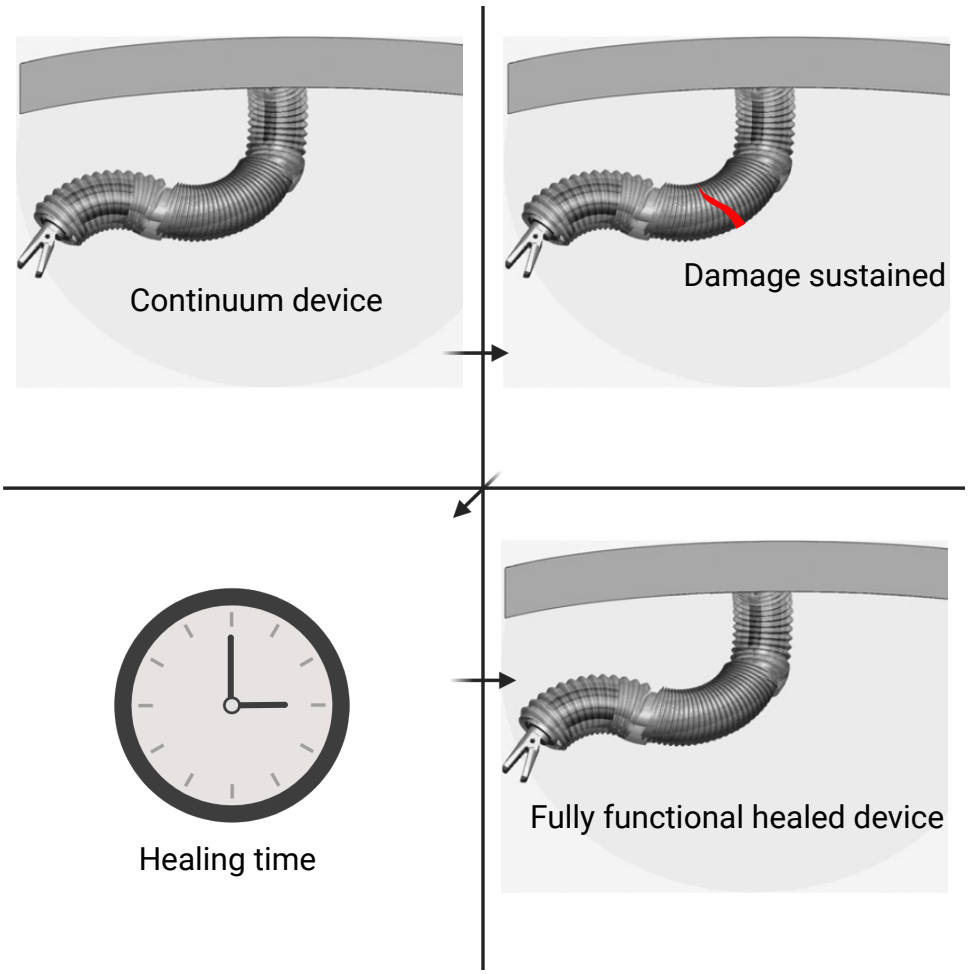


Figure 1.1: A continuum device sustaining damage before self-healing and regaining full functionality using a modified image from[?]

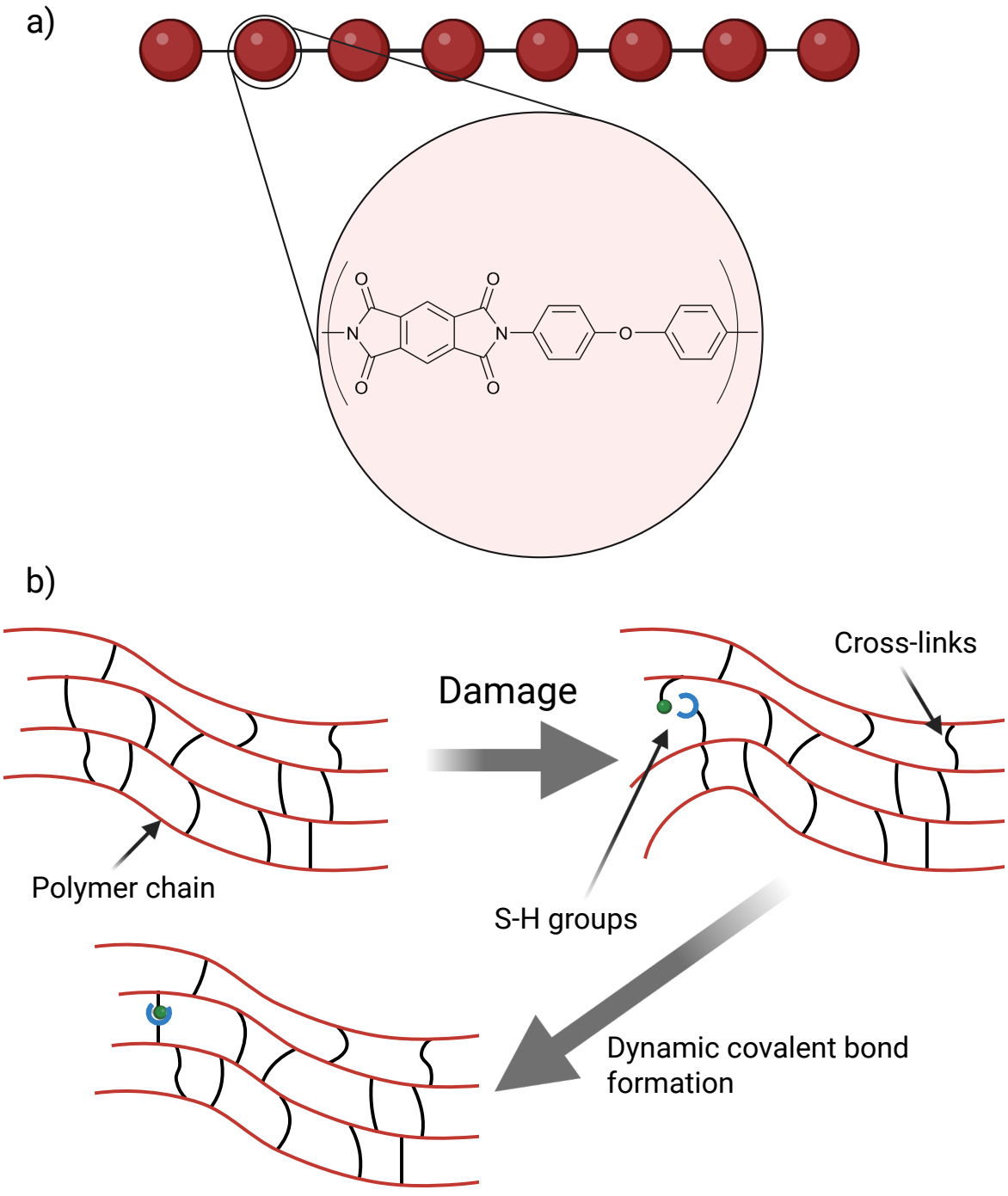


Figure 1.2: a) The polymer chain with a repeating chemical sub-unit. b) The mechanism of damage and repair for a self-healing polymer

tions between multiple polymer chains to reconnect chains that have been shifted apart through damage. The process of damaging or cutting a polymer does not break bonds in the polymer backbone but instead can break interchain interactions called cross-linking, allowing the chains to drift apart, reforming the cross-linking interactions will "heal" the material.[?]

S-H polymers can be classified into intrinsic or extrinsic healing, and autonomous or non-autonomous.[?] Materials with an intrinsic healing capacity involve a healing mechanism through chemical bonds or interactions that are continuously present in the polymer, rather than extrinsic healing, such as capsules of "healing" material that are broken through damage and causing the material to reform.[?] Autonomy refers to the healing process occurring without external input. For example, capsules that burst upon damage of the material and release a healing agent are classified as autonomous, whereas a photochemical reaction that needs UV light to activate[?,?] is a non-autonomous mechanism.

1.3 Comparison of Different Self-Healing Pathways

There are a variety of different pathways that can be taken to create a S-H polymer depending on the chemistry of the initial polymer. The pathways that will be focused on are intrinsically self-healed types, however, both autonomous and non-autonomous types will be considered.

The three pathways below rely on a property of some electrons called conjugation. Conjugation occurs when specific electron orbitals overlap in such as way that allows the electrons to move around the molecule. This is often exploited with a property of some atoms called electronegativity, having a highly electronegative atom in a molecule will attract conjugated electrons to create a local high density region of electrons and therefore a polarisation in a molecule.

Dynamic covalent bonds can form and break based on the conditions applied to them[?] and they are ideal for use as mechanisms for S-H due to their intrinsic nature and reliability in production of high yields.[?] This report investigates the use of two types of dynamic covalent bonding, Diels-Alder and Michael additions, and an example of a supramolecular interaction. Supramolecular chemistry is based on interactions that are non-covalent and may be more suited to a soft material.[?] This is due to the S-H mechanism of the cross-links that forms interactions therefore the polymer chains can move semi-independently.[?,?] Whereas, a dynamic covalent bond has a much higher bond energy[?] and can therefore get closer to the original physical properties of a polymer.

1.3.1 Using a Diels-Alder Addition as a Self-Healing Pathway

Diels-Alder (DA) reactions require a diene, a class of molecule characterised by having two sets of double bonds, and an alkene ("dieneophile").[?,?] The reaction occurs when conjugated electrons from the diene are free to move to form new bonds with the carbons either side of the alkene double bond, this creates a new ring structure. A faster reaction is generally observed with an electron rich diene and an electron poor dienophile.[?] The electronic character of the reagents can be altered by changing their local substituents. This class of reactions can have lots of different conditions depending on the molecules involved. Some variations can occur at room temperature,[?] which may make them an ideal choice for an autonomously healed medical device as the healing can occur whilst deployed in the body. However, many DA reactions are light[?,?] or heat activated.[?]

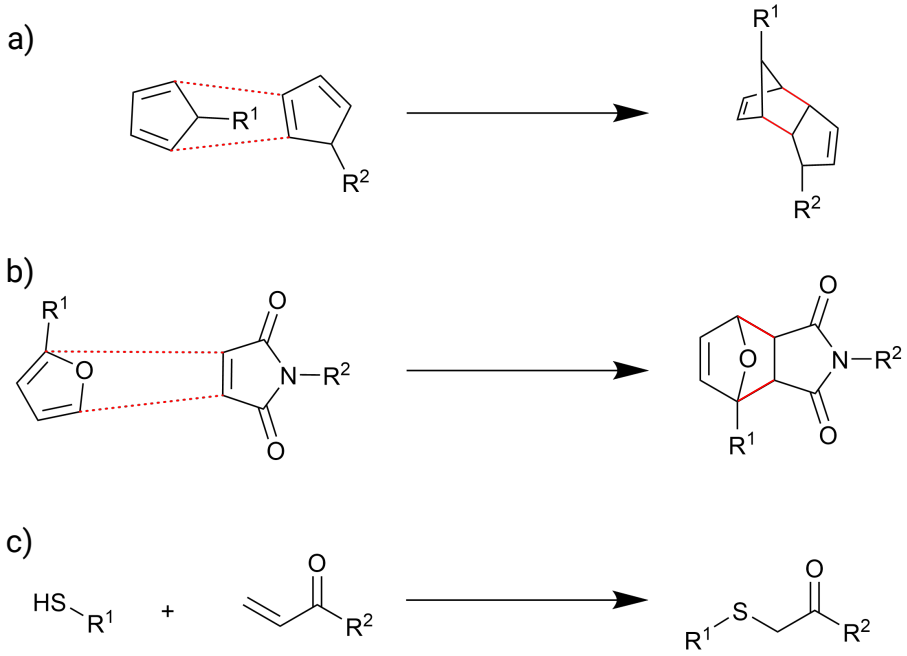


Figure 1.3: a) A Diels-Alder reaction between two molecules of cylopentadiene. b) A Diels-Alder reaction between a furan and maleimide. c) A Thiol-Michael Addition between a thiol and a conjugated alkene.

Some molecules such as cyclopentadiene can perform a DA reaction with itself, this has the potential to be an ideal choice for a S-H polymer as the molecule will never be in close proximity to a molecule that it cannot react with. However, some evidence has shown that the bonds formed are not very strong and therefore high tensile strength materials may not be possible.[?]

One of the most common types of DA reaction is a heat activated reaction between a derivative of both furan and maleimide.[?] They react extremely accurately and require mild conditions,[?] therefore they are very suitable for application as a robotics material. One of the reasons that this reaction is so effective is that the maleimide has electron withdrawing carbonyl groups which make the dieneophile electron poor and therefore reacting faster. However, this mechanism has a disadvantage compared to cyclopentadiene as there are two halves of the end product. Therefore, there must be a two different processes occurring to create both the furan substituted and maleimide substituted chains. This increases the number of total steps in the reaction which increases the risk of low reaction yields and wasted material.

1.3.2 Using a Thiol-Michael Addition as a Self-Healing Pathway

Thiol Michael additions are a type of reaction that relies on a thiol, a molecule characterised by a sulfur containing group and a conjugated alkene.[?] The conjugated alkene used in a Michael addition is with a carbonyl group that will attract the electrons from the alkene to allow the thiol group to react. This has a similar advantage to the Diels-Alder reaction that it is not totally dependant on specific reactants merely groups of reactants. Therefore slight changes can be made to the adjoining groups to tweak the reaction conditions. This has been previously shown to have been successful with a heat

activated reaction that could heal scratches after 15 to 30 minutes at 60 C.[?]

1.3.3 Using a Pi-Stacking Approach as a Self-Healing Pathway

Pi-Stacking is a type of intermolecular interaction whereby conjugated systems interact based on the neighbouring substituents determining if a region is electron rich or poor. These electron disparities cause an electrostatic interaction. This method commonly uses pyrenyl groups, these are polycyclic aromatic compounds, a very pi-electron rich species. These can be interacted with a napthalene diimide complex, an electron poor pi-system to give four pi-stacking interactions. A method has shown the creation of a polymer chain end group which can sandwich a napthalene group within two pyrenyl groups.[?] This method can be combined with other intermolecular interactions like hydrogen bonding[?] or mettalosupramolecular interactions to give a stronger interaction.[?]

1.4 Applications of Self-Healing Polymers in Soft Robotics

Continuing the theme of biological inspiration of soft robotics, S-H polymers can be used to extend the lifespan of components. In soft continuum robots, the goal is usually to make the robot softer than the environment surrounding it.[?] This presents a challenge that the environment can often be abrasive and can damage the device. It is therefore evident that a damage resilient material would be useful. Damage resilient materials can take multiple forms: Damage isolation, material elasticity, physical self healing such as low melting temperature materials and chemical self healing.[?]

1.4.1 State of the Art Self-Healing Soft Robots

State-of-the-art approaches using intrinsic S-H mechanisms are commonly based on DA chemistry.[?] These have been successfully implemented in tendon-driven[?] and pneumatically controlled soft robotic devices.[?,?] Such devices are capable of recovering from macroscopic damage, including punctures and cuts. Additionally, there are non-covalent approaches that utilize hydrogen bonding in the development of soft self-healing actuators, such as in McKibben muscle designs.[?]

1.4.2 Requirements for Self-Healing Polymers in Soft Robotics

Terryn *et al.*[?] states that there are 5 main criteria for a S-H polymer; macroscopic damage must be healed completely; healing can take place multiple times at the same location; there are minimal material property changes after healing; elastomeric behaviour is achieved; the reprocessing and recycling of a material is possible. Here we propose that elastomeric behaviours are non-essential and a more useful requirement would involve a timely S-H reaction.

1.5 Limitations with Current Materials for Soft Robotics

Damage control whilst a device is active can be a challenge,[?] this shows a need for autonomous S-H.[?] Non-autonomous S-H can be made "functionally autonomous" through

the use of stimuli responsive mechanisms, however, the current options are slow in operation.[?] Currently recovery timescales of applied S-H polymers are around 25 hours.[?,?] By altering the polymer used or S-H pathway this could be improved. There is also a lack of an easy synthesis route for many of the current polymer options,[?] thereby increasing the cost and reducing widespread use. Also the number of damage-repair cycles in these devices are low, such as 3 or less.[?] Most devices do not achieve all 5 criteria laid out in the Terryn *et al.* review,[?] such as the McKibben design of A. Lopez Diaz *et al.*[?] only reports a healing efficiency of 65%.

Chapter 2

Project Aims and Objectives

Chapter 3

Synthesis of a Self-Healing Polymer

3.1 Results

3.1.1 Hydrolysis of the Polyimide

The hydrolysis of the polyimide follows a standard mechanism for hydrolysis of an amide to a carboxylic acid using sodium hydroxide (NaOH). These show the hydroxy group attacking the carbonyl which then breaks the C-N bond and gives the carboxylic acid free.

Structural Analysis

Structural determination was conducted via Infrared spectroscopy. A comparison was made between the attempted hydrolysis of the sheets and a purchased form (polyamic acid).

The infrared spectrum of the hydrolysis attempts indicate that the crosslinking was not a successful reaction as the peaks in the spectrum are extremely similar to the unreacted form of the polymer. Also, in a comparison to the purchased PAA, which should be the hydrolysed product that we were hoping for, the region around 3000 cm^{-1} is much busier and has the characteristic wide OH region. This would indicate that the hydrolysis hasn't occurred. This maybe due to the imide ring breaking and reforming as the temperature of the reaction is likely higher than the imide formation temperature meaning that the amide bond may get broken but in fact reforms at the high temperature. This seems a little unlikely though as I think there would be a suspension of material along with the main polymer.

The amount of NaOH was chosen to be in the ratio of 1:4 NaOH: monomer as there was a concern that we may fully hydrolyse the polymer (at all 4 amide bonds in the two imide rings) and give a tetrasubsituted benzene ring.

3.1.2 Cross-linking of the Polyimide

Cross-linking attempts follow a mechanism as proposed by (put reference here), where ethylene carbonate is used to form an ester with the carboxylic acids on neighbouring polymer chains, thereby joining them together with a "cross-link". This was performed with three different salts (TEAB, $ZnCl_2$, $FeCl_3$) as catalysts.

During the cross-linking reaction with TEAB, an orange-brown suspension was formed that when washed, filtered and freeze dried gave a pale orange powder. With $ZnCl_2$, the

liquid was a dark orange clear liquid that, when water was added, gave a single clump of grey polymeric residue. This was freeze dried to remove the excess water and then analysed. The cross-linking reaction with the $FeCl_3$, gave a dark orange/brown gel, which gave a grey gum-like texture in small pieces when water was added. This was also frozen and analysed.

During a cross-linking with ethylene carbonate, CO2 is released. This was a concern as the monowave reactor has a pressure limit of 20 bar. To ensure there was a reliable safety buffer, we set an artificial limit of 5 bar. The calculation to ensure this limit was kept used PV = nRT. (insert calculations).

From the IR data, the crosslinking with TEAB was unsuccessful as the product didnt have the key peaks of the C-O bond in the ester or any alcohol peak. The alcohol peaks would be likely as the crosslinking was unlikely to be 100% successful and there would likely be the single ester product formed. However, neither of these were present and since the product was insoluble, no NMR could be obtained. This IR specturm was compared with the spectrum of Kapton (i.e. the ring closed form of the polyamic acid) and the similarities confirmed that the imide ring reformation had occured. The absorbance of the synthsised form from the PAA has much stronger absorbances. THis is likely due to the fact that the clamping force on the kapton film is not as strong as in the powder and the powdered material is thicker so will absorb more just due to the material.

From the IR data of the ZnCl2 and the FeCl3, there are two new peaks at around 1800 which are not in either the kapton or the polyamic acid. These are likely the

3.2 Hydrolysis of a Polyimide

The hydrolysis of the poylimide (kapton) is needed to open an amide bond on the imide ring to reveal a carrboxylic acid group whilst the bolymer back bone is still intact. This will make the polymer soluble, however since the polymer is insoluble currently the only attack will be on the polymer surface. This was completed with sodium hydroxide in (solvent) and in the monowave reactor for (x time)

The structure of the proposed polyimide is shown in figure (x). This structure has multiple imide rings that we hypothesise are a good starting point for introducing reactivity.

The partial hydrolysis of these rings was performed with Sodium hydroxide solution (in dmso). This reaction was performed in the monowave reactor. The monowave reactor is a reaction vessel that is within a machine that can reduce the pressure on the environment. This is useful as the boiling point of the solvent is increase with lower pressure. Therefore a reaction that can only be performed above a certain temperature like this one can be done in solvent with a lower boiling point than the reactuion requirements.

3.2.1

3.2.2 Comparison of Hydrolysed Polyimide versus Purchased Polyamic Acid

the hydrolysed polyimide should have a similar ir as a purchased form of the polyimide as both have a exposed carboxylic acid group, therefore there will be a characteristic shift at (wavenumber).

3.2.3 Spectral Analysis

Structural determination was performed with infrared (IR) spectroscopy and Ultraviolet - visible (UV-Vis) spectroscopy. Key IR peaks for the unmodified Kapton are (x), the key peask for the modified kapton are (x) and the key peaks for the purchased Polyamic acid are (y).

3.3 Crosslinking of Polyamic Acid

crosslinking of polyamic acid was done in a reaction with the ethylene carbonate and an ammonium bromide salt this gave the reaction as in a similar process as in (this paper). the carboxylic acid groups react with the ethylene carbonate to form either an ester with a hydroxy after an alkyl chain on one of the polymer chains or it will like the polymer chains together by forming a diester.

Using the monowave uses a 10mL reaction tube that was filled with 4ml of reagents to leave room for the co2 that is released in the reaction. The volume of co2 released was calculated using $PV = nRT$ which was rearranged to $n = \frac{PV}{RT}$. The monowave 50 has a maximum pressure of 20 bar, so we opted for a maximum of 5 bar for safety. this was combined with a temperature of 150C to give:

$$n = \frac{PV}{RT}$$
$$n =$$

THis was the total number of co2 mo

3.4 Addition of Diels-Alder Groups

3.5 Experimental

Kapton (poly-oxydiphenylene-pyromellitimide) sheets were cut into small pieces ($\approx 2\text{mm}^2$). These were added to a solution of NaOH (vol and mol here) and heated to 120 C in a monowave reactor set to ramp for 20 mins and hold for 20 and 40 mins successively. The reaction gave no observable difference in the Kapton sheet

3.6 Experimental

3.6.1 The Hydrolysis of a Polyimide

Poly..... (Kapton) film (0.2g, thickness of x) was cut into pieces of approx. 3mm^2, these were mixed with a solution of NaOH () in a reaction tube. This tube was placed into a monowave reactor that was set to ramp for 20 minutes and hold for 20 minutes at 100 C. A sample was taken and the remaining mixture was reacted again on the monowave for ramp 20 mins and hold 20 minutes at 100C. There was no observable difference in the Kapton film. Then this was thoroughly washed with water and IR was collected.

3.6.2 Crosslinking of Polyamic acid

This procedure outlines the crosslinking procedure of Polyamic acid via ethylene carbonate using a catalytic salt as outlined in previously published procedures.

3.6.3 Cross-linking with TEAB and Ethylene carbonate

A mixture of polyamic acid (0.51mL, conc) was mixed with ethylene carbonate (0.08g) and tetraethylammonium bromide (). This was mixed in a monowave reactor set to ramp for 20 minutes and hold for 1 hour at 150 C. This gave an orange suspension that was dried to give a pale orange powder as the final product (grams, yield).

3.6.4 Cross-linking with other salts

The crosslinking with ZnCl2 and FeCl3 was performed at 100C for 1 hour. With ZnCl2 a clear dark brown viscous liquid was obtained and with the FeCl3, a clear dark brown gelatinous liquid was obtained. A solid was extracted from each of these liquids and spectroscopic analyses were performed.

3.6.5 Spectral Data

tetraethylammonium bromide Crosslinking

A orange powder (grams,moles,yield) was isolated, ν max/cm^{-1}(solid);

Iron chloride Crosslinking

A white gum (grams,moles,yield) was isolated, δ H(400MHz, DMSO); , δ C(100MHz, DMSO); , ν max/cm^{-1}(solid);

Zinc chloride Crosslinking

Brittle brown pieces (grams,moles,yield) were isolated, δ H(400MHz, DMSO); , δ C(100MHz, DMSO); , ν max/cm^{-1}(solid);

Chapter 4

Material Comparison